

Troubleshooting Execution Report — Case 2031442977

To: Mai Ezz Eldin — Lenovo Premium Care **Re:** Case 2031442977 — Legion Pro 5 16IRX10 (MT 83NN) — VIDEO_TDR_FAILURE **From:** Ruslan Abuzant **Date:** 17 June 2026

Dear Mai,

Thank you again for escalating this to Tier 3 Engineering and Product Quality. As promised, I have executed all of the troubleshooting steps your team requested. This report documents each step, the result, and the outcome of the monitoring period.

I want to be clear up front about the headline result, because it is decisive:

I completed every requested step — including installing Lenovo's own validated OEM driver — and the machine crashed with the identical signature 25 minutes into idle. The defect is confirmed to be independent of driver version and independent of every setting your team recommended.

The detail follows.

Per-Step Execution Log

Step 1 — Clean driver reinstallation via DDU, using the Lenovo-validated OEM driver

- **Status:**  Completed 17 June 2026 (second documented clean install of this driver)
- **Action:** Disconnected from network, booted into Safe Mode, ran Display Driver Uninstaller (NVIDIA, clean + no restart), then installed the Lenovo-validated OEM driver `NV_VGA_32.0.15.8205_WHQL` downloaded directly from the official 83NN support page.
- **Verified post-reboot:** Driver version `32.0.15.8205` confirmed active.
- **Important note for engineering:** This is the **second** clean DDU install of this exact OEM driver during the investigation. The crash also occurred on this OEM driver during the first install earlier in the investigation, which is why the current NVIDIA Studio driver (596.36 / `32.0.15.9636`) was subsequently tested.

The identical crash signature has now been recorded on **all three of the following drivers**:

Driver	Source	Result
Factory-shipped driver	Lenovo (as delivered)	Crashed (6 crashes, Jan-May 2026)
OEM driver (32.0.15.8205)	Lenovo support site	Crashed (twice — including today)
Studio driver 596.36 (32.0.15.9636)	NVIDIA (current)	Crashed

There is no driver version — OEM or current — on which this defect does not reproduce.

Step 2a — Disable Hardware-Accelerated GPU Scheduling (HAGS)

- **Status:**  Completed 17 June 2026
- **Action:** Set (HKLM\SYSTEM\CurrentControlSet\Control\GraphicsDrivers\HwSchMode = 1) (disabled). Confirmed via registry and Windows Settings UI.
- **Verified:** HAGS remained disabled ((HwSchMode = 1)) after the driver reinstall and reboot. This was the only genuinely new variable in the troubleshooting batch. **It did not prevent the crash.**

Step 2b — Set TdrDelay = 10

- **Status:**  Already in place prior to your request
- **Note:** Both (TdrDelay) and (TdrDDIDelay) were already set to (10) (decimal) before this case. **The crashes have been occurring despite a recovery timeout already extended to 5× the 2-second default.** This is diagnostically significant: it indicates the GPU is **hard-hanging**, not merely responding slowly. Granting the GPU five times the normal recovery window did not allow it to recover. A masking parameter cannot fix a hard hang, and this confirms it.

Step 3 — BIOS update

- **Status:**  Already at latest available
- **Note:** BIOS (S9CN18WW) (24 Nov 2025) is the latest version published for MT 83NN as of today. No newer BIOS exists. I additionally updated the Intel ME firmware to the latest ((16.1.40.2765)) during the prior investigation. There is no firmware update available to me that has not already been applied.

Step 3 — Lenovo Vantage thermal & GPU mode

- **Status:**  Completed 17 June 2026

- **Action:** Thermal Mode set to **Balanced** (per your "Balance or Performance" instruction). GPU Working Mode confirmed as **Hybrid**.
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Monitoring Result

With **every step above applied and verified** — Lenovo OEM driver (8205), HAGS disabled, TdrDelay at 10, Balanced thermal, Hybrid GPU mode — I placed the machine into normal idle to monitor for recurrence.

- **Monitoring began:** 17 June 2026, approximately 18:20
- **Crash occurred:** 17 June 2026, **18:44:37** — approximately **25 minutes** after going idle
- **Dump file:** (061726-14625-01.dmp)

Crash Signature (identical to all prior crashes)

Field	Value
Bug check	(0x116) VIDEO_TDR_FAILURE
Faulting offset	(dxgkrnl.sys + 186d5d) (identical to every prior crash)
NTSTATUS	(0xC000009A) (STATUS_INSUFFICIENT_RESOURCES)
Responsible module	(nvlddmkm.sys) (Lenovo OEM driver 8205)

This is the same (dxgkrnl) faulting offset, the same NTSTATUS, and the same failure pattern recorded in every previous crash on this machine and on the prior unit — now reproduced on Lenovo's own validated OEM driver with Lenovo's own recommended settings.

What This Result Establishes

The troubleshooting batch your team prescribed has now been executed in full and has **failed to resolve the issue**. Specifically, this result eliminates the remaining hypotheses:

1. **It is not a driver-version issue.** The crash occurs on the Lenovo factory driver, the Lenovo OEM driver (8205) (twice), and the current NVIDIA Studio driver. Every driver tested — including the one Lenovo validates and recommends — reproduces it.
2. **It is not resolved by disabling HAGS.** The only new variable in the batch did not prevent the crash.

3. **It is not a "slow GPU recovery" issue that TdrDelay can mask.** The GPU fails to recover even with a 5× extended timeout, confirming a hard hang.
4. **It is not a firmware-version issue addressable by the customer.** BIOS and ME firmware are both at the latest available versions.
5. **It is not an isolated hardware defect.** The identical signature has now appeared across three physical units of this model.

Additional Engineering Evidence

For your Tier 3 team's benefit, I have independently isolated the crash mechanism through controlled testing:

- The crash occurs when the **dGPU is permitted to enter deep idle**.
- The crash does **not** occur when the dGPU is kept continuously engaged with a hardware-accelerated workload.
- Most precise data: **19 days 22 hours of continuous uptime** with the GPU kept busy, versus **6 crashes in the 14 days immediately prior** without that workload, and a crash **within 24 hours** of inadvertently removing the workaround after a reboot.

This points the root cause to the **dGPU's deep-idle entry/exit power-management path** on the RTX 5060 Laptop (Blackwell mobile) as integrated on this platform — a driver/VBIOS/EC-firmware interaction, not a user-configurable setting.

NVIDIA's engineering team has independently accepted a bug report on this issue and is reviewing the full 4 GB kernel dump. I am happy to coordinate so that Lenovo and NVIDIA engineering can share findings directly.

My Position and Request

I have now cooperated fully and exhaustively. I have executed every step your team requested, documented each one, used Lenovo's own validated driver, and the defect persists with an identical signature. There are no further customer-side actions available that have not already been performed and documented.

Given that:

- The issue is confirmed systemic across three units,
- It persists on Lenovo's own validated OEM driver and Lenovo's own recommended settings,
- It is not addressable by any firmware or configuration available to me,
- And the root cause has been isolated to the GPU deep-idle path at the driver/firmware level,

I am requesting that Lenovo now commit to one of the following:

1. **An engineering resolution with a timeline** — a BIOS/EC-firmware update, a coordinated VBIOS/driver fix with NVIDIA, or equivalent — that addresses the deep-idle defect at its source; **or**
2. **A replacement with a different (non-affected) model, or a buyback/refund of this unit**, given that three consecutive units of this model have failed identically and a fourth unit of the same model would be expected to fail the same way.

I remain willing to accommodate the scheduled on-site technician visit as a cooperative step, with the understanding (as previously noted) that any hardware change must be monitored for a minimum of 7 days before being considered a resolution, since the crash occurs only at idle and may take up to several days to recur.

The complete evidence package — all crash dumps (eight-plus across two units, including the 4 GB full kernel dump), full WinDbg analyses, configuration documentation, and the controlled reproduction data — remains available to your engineering team on request.

Thank you for your continued engagement on this. I would appreciate confirmation of the path forward from Tier 3 Engineering.

Best regards, Ruslan Abuzant Case 2031442977

Attachments available on request: crash dump files, WinDbg (!analyze -v) outputs, powercfg energy report, system configuration export.